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Math 208 Final Exam Term 251

Original exam date: 22 December 2025. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The general solution of the system $X' = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix} X$ is given by

(a) $X(t) = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1+t \\ t \end{bmatrix} e^{2t}$

(b) $X(t) = c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} -1+t \\ t \end{bmatrix} e^{2t}$

(c) $X(t) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1+t \\ t \end{bmatrix} e^{2t}$

(d) $X(t) = c_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1+t \\ 2t \end{bmatrix} e^{2t}$

(e) $X(t) = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 2+t \\ t \end{bmatrix} e^{2t}$

2. By using the method of undetermined coefficients, a particular solution of the differential equation

$$y'' + 4y' + 3y = 8e^{x} + 3e^{-2x}$$

is given by

(a) $y_p = e^{x} - e^{-2x}$

(b) $y_p = e^{x} + 2e^{-2x}$

(c) $y_p = 2e^{x} + e^{-2x}$

(d) $y_p = e^{x} - 3e^{-2x}$

(e) $y_p = 3e^{x} - 2e^{-2x}$

3. The general solution of $X' = \begin{bmatrix} 0 & 3 & 1 \\ 1 & 2 & 1 \\ 1 & 3 & 0 \end{bmatrix} X$ can be written as

$$X = c_1 \begin{bmatrix} 1 \\ 1 \\ \alpha \end{bmatrix} e^{4t} + c_2 \begin{bmatrix} 3 \\ \beta \\ 0 \end{bmatrix} e^{-t} + c_3 \begin{bmatrix} \gamma \\ 0 \\ -1 \end{bmatrix} e^{-t},$$

then $\alpha + \beta + \gamma =$

(a) 4

(b) 1

(c) 3

(d) 0

(e) -1

4. If $c_0 \neq 0$,

$$c_k = \frac{\left(k - \frac{5}{2}\right)\left(k + \frac{3}{2}\right)}{2k(2k+1)}\, c_{k-1}, \quad k \geq 1,$$

is the recurrence relation corresponding to the indicial root $r = \dfrac{1}{2}$ in the series solution of the differential equation $x(4-x)y'' + (2-x)y' + 4y = 0$ about $x = 0$, then the solution is given by

(a) $y = x^{\frac{1}{2}}\left[1 + \dfrac{3}{8}x + \dfrac{7}{128}x^2 + \ldots\right]$

(b) $y = x^{\frac{1}{2}}\left[1 + \dfrac{5}{8}x + \dfrac{7}{128}x^2 + \ldots\right]$

(c) $y = x^{\frac{1}{2}}\left[1 - \dfrac{3}{8}x + \dfrac{7}{64}x^2 + \ldots\right]$

(d) $y = x^{\frac{1}{2}}\left[1 - \dfrac{5}{8}x - \dfrac{7}{128}x^2 + \ldots\right]$

(e) $y = x^{\frac{1}{2}}\left[1 - \dfrac{5}{8}x + \dfrac{7}{128}x^2 + \ldots\right]$

5. The general solution of the linear differential equation

$$x \ln x \frac{dy}{dx} - y = 3x^3(\ln x)^2, \qquad x > 1,$$

is given by

(a) $y = (c + x)\ln x$

(b) $y = \left(c + x^4\right)\ln x$

(c) $y = \left(c + x^2\right)\ln x$

(d) $y = (c + 3x)\ln x$

(e) $y = \left(c + x^3\right)\ln x$

6. Consider the non-homogeneous system $X' = AX + \begin{bmatrix} 4 \\ -1 \end{bmatrix}$. If the general solution of the associated homogeneous system is

$$X_c = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^t,$$

then the value of the particular solution $X_p(-1) =$

(a) $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$

(c) $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$

(d) $\begin{bmatrix} -4 \\ 1 \end{bmatrix}$

(e) $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$

7. For the matrix $A = \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}$, if P is a diagonalizing matrix such that $P^{-1}AP = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$, then $P =$

(a) $P = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$

(c) $P = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$

(d) $P = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$

(e) $P = \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$

8. The general solution of the differential equation $y^{(6)} + 2y^{(4)} + y'' = 0$ is given by

(a) $y(x) = c_1 + c_2x + c_3 \cos x + c_4 \sin x + c_5 \cos(2x) + c_6 \sin(2x)$

(b) $y(x) = c_1 + c_2x + c_3e^{-x} + c_4e^x + c_5 \cos x + c_6 \sin x$

(c) $y(x) = c_1 + c_2x + (c_3 + c_4x)e^x \cos x + (c_5 + c_6x)e^x \sin x$

(d) $y(x) = c_1 + c_2e^x + c_3e^{-x} + c_4 \cos x + c_5 \sin x + c_6x \sin x$

(e) $y(x) = c_1 + c_2x + (c_3 + c_4x) \cos x + (c_5 + c_6x) \sin x$

9. Consider the system $X' = AX$, and $X(0) = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$, where A is a 2×2 matrix with real entries. If A has an eigenvalue $\lambda = 2 + 2i$ with corresponding eigenvector $K = \begin{bmatrix} -5 \\ 1 + 2i \end{bmatrix}$, then $X\left(\frac{\pi}{2}\right) =$

(a) $\begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{\pi}$

(b) $\begin{bmatrix} 3 \\ 0 \end{bmatrix} e^{\pi}$

(c) $\begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{\pi}$

(d) $\begin{bmatrix} 5 \\ -3 \end{bmatrix} e^{\pi}$

(e) $\begin{bmatrix} 0 \\ 3 \end{bmatrix} e^{\pi}$

10. The three vectors $\mathbf{v}_1 = (2, -1, 4)$, $\mathbf{v}_2 = (0, 5, 1)$ and $\mathbf{v}_3 = (5, 0, B)$ of $\mathbb{R}^3$ are linearly dependent if $B =$

(a) 5.5

(b) 10.5

(c) 8.5

(d) 7.5

(e) 6.5

11. The general solution of the homogeneous differential equation

$$y(x + y)\,dx - x^2\,dy = 0$$

is

(a) $y \ln |y| + x^2 = cy$

(b) $y \ln |y| + x = cy$

(c) $x \ln |y| + y = cx$

(d) $y \ln |x| + x = cy$

(e) $x \ln |x| + x^2 = cy$

12. The minimum radius of convergence of a power series solution of the differential equation $\left(x^2 - 2x + 5\right) y'' + xy' - y = 0$ about the ordinary point $x = -2$ is equal to

(a) $2\sqrt{2}$

(b) $\sqrt{11}$

(c) 0

(d) ∞

(e) $\sqrt{13}$

13. A possible fundamental matrix $\Phi(t)$ of the system $X' = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} X$ is

(a) $\begin{bmatrix} e^t & e^{-t} \\ e^t & 2e^{-t} \end{bmatrix}$

(b) $\begin{bmatrix} 3e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix}$

(c) $\begin{bmatrix} e^t & e^{-t} \\ 3e^t & e^{-t} \end{bmatrix}$

(d) $\begin{bmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix}$

(e) $\begin{bmatrix} 2e^t & e^{-t} \\ e^t & 3e^{-t} \end{bmatrix}$

14. The general solution of the exact differential equation

$$\left(y \sec^2(xy) + \sin x\right) dx + \left(x \sec^2(xy) + \sin y\right) dy = 0$$

is given by

(a) $\tan(xy) - \cos x - \cos y = c$

(b) $\tan(xy) + \cos x - 2\cos y = c$

(c) $x \tan(xy) - \cos x - \cos y = c$

(d) $y \tan(xy) + \cos x + \cos y = c$

(e) $2 \tan(xy) - \cos x + \cos y = c$

15. If $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ is a series solution for the differential equation $2xy'' - y' + 2y = 0$ about $x = 0$, then the non-integer indicial root is equal to

(a) $\frac{3}{4}$

(b) $\frac{2}{3}$

(c) $\frac{1}{2}$

(d) $\frac{3}{2}$

(e) $\frac{4}{3}$

16. If $X = \begin{bmatrix} a \\ 2 \end{bmatrix} e^{-3t/2}$ is a solution of the system $X' = \begin{bmatrix} -1 & \frac{1}{4} \\ 1 & -1 \end{bmatrix} X$, then $a =$

(a) 1

(b) 0

(c) -1

(d) -2

(e) 2

17. The rank of the matrix

$$\begin{bmatrix} 1 & 4 & 5 & 2 \\ -2 & -8 & -10 & -4 \\ 3 & 12 & 15 & 6 \\ 0 & 0 & 3 & 0 \end{bmatrix}$$

is

(a) 1

(b) 2

(c) 0

(d) 4

(e) 3

18. If $K = \begin{bmatrix} 1 \\ a \\ -13 \end{bmatrix}$ is an eigenvector with eigenvalue $\lambda = 0$ of $A = \begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix}$, then $a =$

(a) 7

(b) 4

(c) 6

(d) 3

(e) 5

19. If $y = \sum_{n=0}^{\infty} c_n x^n$ is a power series solution of the differential equation $(2x+1)y'' + y' = 0$ about the ordinary point $x = 0$, then the constants c_n are given according to the recurrence relation:

(a) $c_n = -\dfrac{3n-2}{n}\, c_{n-1},\ n \geq 1$

(b) $c_n = -\dfrac{3n-2}{n}\, c_{n-1},\ n \geq 2$

(c) $c_n = -\dfrac{2n-3}{n}\, c_{n-1},\ n \geq 2$

(d) $c_n = \dfrac{3n-1}{n}\, c_{n-1},\ n \geq 2$

(e) $c_n = \dfrac{2n+3}{n}\, c_{n-1},\ n \geq 1$

20. If $A = \begin{bmatrix} 3 & 4 \\ 0 & 3 \end{bmatrix}$, then $e^{At} =$

(a) $\begin{bmatrix} e^{3t} & 4te^{3t} \\ 0 & e^{3t} \end{bmatrix}$

(b) $\begin{bmatrix} e^{3t} & 4e^{3t} \\ 0 & e^{3t} \end{bmatrix}$

(c) $\begin{bmatrix} e^{3t} & e^{4t} \\ 0 & e^{3t} \end{bmatrix}$

(d) $\begin{bmatrix} 3e^{3t} & 4te^{t} \\ 0 & 3e^{3t} \end{bmatrix}$

(e) $\begin{bmatrix} e^{3t} & 3te^{3t} \\ 0 & e^{3t} \end{bmatrix}$

21. A particular solution of the differential equation

$$y'' + y = \sec x \tan x$$

is given by $y_p(x) =$

(a) $3x \cos x + \sin x \ln|\tan x|$

(b) $2x \cos x + \sin x \ln|\sec x|$

(c) $2x \cos x + \sin x \tan x$

(d) $x \cos x + \sin x \ln|\cos x|$

(e) $x \cos x - \sin x + \sin x \ln|\sec x|$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$(\lambda-2)^2=0$, $\mathbf{v}_1=(1,1)$, $\mathbf{v}_2=(1,0)$: $X_2=(1+t,t)e^{2t}$.
2	**(d)**	$y_p=Ae^{x}+Be^{-2x}$: $8A=8$, $-B=3$, so $y_p=e^{x}-3e^{-2x}$.
3	**(b)**	$\lambda=4$: $(1,1,1)$; $\lambda=-1$: $x+3y+z=0$ gives $\beta=-1$, $\gamma=1$; sum 1.
4	**(e)**	$c_0=1$: $c_1=\frac{(-3/2)(5/2)}{6}=-\frac{5}{8}$, $c_2=-\frac{7}{80}c_1=\frac{7}{128}$.
5	**(e)**	$P=-\frac{1}{x\ln x}$, $\rho=\frac{1}{\ln x}$: $(\frac{y}{\ln x})'=3x^2$, so $y=(c+x^3)\ln x$.
6	**(d)**	$\Phi^{-1}F=(-11,5e^{-t})$, $X_p=(-11t-15,\,-11t-10)$, so $X_p(-1)=(-4,1)$.
7	**(b)**	$\lambda=2$: $(1,1)$; $\lambda=5$: $(-1,2)$; columns in that order.
8	**(e)**	$r^2(r^2+1)^2=0$: $r=0,0$ and $r=\pm i$ (double).
9	**(d)**	$\mathbf{a}=(-5,1)$, $\mathbf{b}=(0,2)$, $c_1=c_2=1$; at $t=\frac{\pi}{2}$: $-e^{\pi}(\mathbf{a}+\mathbf{b})=(5,-3)e^{\pi}$.
10	**(b)**	$\det=10B-105=0$, so $B=10.5$.
11	**(d)**	$v=y/x$: $xv'=v^2$, $-\frac{x}{y}=\ln\lvert x\rvert+C$, so $y\ln\lvert x\rvert+x=cy$.
12	**(e)**	Singular points $1\pm 2i$; distance from -2 is $\sqrt{3^2+2^2}=\sqrt{13}$.
13	**(d)**	$\lambda=1$: $(1,1)$; $\lambda=-1$: $(1,3)$; columns $\mathbf{v}_ie^{\lambda_i t}$.
14	**(a)**	$F=\int M\,dx=\tan(xy)-\cos x+g(y)$, $g=-\cos y$: $\tan(xy)-\cos x-\cos y=c$.
15	**(d)**	$p_0=-\frac{1}{2}$, $q_0=0$: $2r^2-3r=0$, so $r=0,\frac{3}{2}$.
16	**(c)**	$A\mathbf{v}=-\frac{3}{2}\mathbf{v}$: second row $a-2=-3$, so $a=-1$.
17	**(b)**	Rows 2, 3 are multiples of row 1; pivots in $(1,4,5,2)$, $(0,0,3,0)$: rank 2.
18	**(c)**	$AK=(2a-12,\,6-a,\,12-2a)=\mathbf{0}$ gives $a=6$.
19	**(c)**	Coefficient of x^{n-1}: $(2n-1)c_n+(n+1)c_{n+1}=0$, so $c_n=-\frac{2n-3}{n}c_{n-1}$, $n\geq 2$.
20	**(a)**	$A=3I+N$, $N^2=0$: $e^{At}=e^{3t}(I+Nt)$.
21	**(e)**	Var. of parameters: $u_1=x-\tan x$, $u_2=\ln\lvert\sec x\rvert$, $W=1$.